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Classification of Farms for Recommendation of Rice Cultivation Using Naive Bayes and SVM: A Case Study

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12.1 INTRODUCTION

The present human population of the world is about 7.6 billion, and it is estimated that the world population will roughly reach 9.8 billion by the end of 2050. Consequently, the food requirements will certainly rise, and contrary to this, the agricultural land is drastically decreasing every passing day due to the miss management in farming, improper planning and unpredictable weather conditions [1], and urbanization. In contemporary times, precision agriculture has gained huge popularity in the agriculture industry with its optimistic results, and it has helped to a great extent in different fields of agriculture like irregular irrigations, plant disease detections, plant counting, crop recommendations, yield estimation, selective harvesting, pest control, and weed detection, as well as many other spheres of agriculture field.

> Precision Agriculture is a management strategy that gathers, processes and analyzes temporal, spatial and individual data and combines it with other information to support management decisions according to estimated variability for improved resource use efficiency, productivity, quality, profitability and sustainability of agricultural production. [2]

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In order to gain more insight into data retrieved from precision agriculture, machine-learning (ML) techniques are highly recommended and used in this field. With the implementation of ML tools, we can streamline the process and reduce errors in data, predictions are easier, and interpretation of imagery data becomes quick and easy. "Machine learning is a branch of computer science that broadly aims to enable computers to 'learn' without being directly programmed" [3]. "A machine or an intelligent computer program learns and extracts the knowledge from the data; the extracted knowledge then builds a framework that helps in making predictions or intelligent decisions. Therefore, the ML process is divided into three key parts – i.e., data input, model building, and generalization" [1]. "Generalization is the process for predicting the output for the inputs with which the algorithm has not been trained before" [1]. The ML algorithms can be used to build the classifier model for detecting the quality of soil for proper cultivation of crops. In this chapter, we will understand a similar subject as a case study. To implement a predictive model, let us take a dataset of 2,200 instances and modify it according to the rice crop cultivation and implement two ML predictive models – i.e., Naive Bayes and the support vector machine (SVM) for predicting the rice crop cultivation farm. We will also perform a comparative study of these two models based on various accuracy measures.

12.2 LITERATURE SURVEY

Rice, as we know, is one of the most consumable crops across the globe. Farmers face a lot of difficulties while making a decision in order to be productive and achieve a sustainable crop in the changing climate. "Machine Learning techniques can be used to improve prediction of crop yield under different climatic conditions" [4]. A good number of researchers have applied ML algorithms to detect rice diseases. Jian and Wei have applied SVM for the identification of cucumber leaf disease [5]. Their results have shown that the SVM method based on the radial basis function (RBF) kernel function made the best performance for the classification of diseases of cucumbers as compared to the polynomial and sigmoid kernel functions. Sethy, Barpanda, Rath, and Behera have worked on deep feature-based rice leaf disease detection using CNN and SVM [6].

Gandhi, Petkar, and Armstrong have used neural networks to predict the production yield of rice and to investigate the factors that affect the "rice crop yield for various districts of Maharashtra state in India" [7]. The parameters that were considered by these researchers while carrying out their study were "precipitation, minimum temperature, average temperature, maximum temperature and reference crop evapotranspiration, area, production and yield for the Kharif season (June to November) for the years 1998 to 2002." Singha and et al. have employed the "artificial intelligence (AI) technique for rice and potato crop yield prediction model in the region of Tarakeswar block, Hooghly District, West Bengal." The variables that were predominately used in the study were "climatic factors, static soil parameters, available soil nutrient, agricultural practice parameters, farm mechanization, terrain distribution and socio-economic conditions." They have used "artificial neural network, Support vector machine and Deep Neural network" in their study [8]. The Hasan et al. study has introduced a novel CNN architecture that is comparatively small in size and fairly promising in performance to predict rice leaf disease with moderate accuracy and lower time complexity [9].

Patil and Kumar have proposed a multimodality data fusion framework for rice disease detection.

> The Rice-Fusion framework initially extracts the numerical features from agro-meteorological data collected from sensors. Next, it extracts the visual features from the captured rice images. These extracted features are further fused using a concatenation layer followed by a dense layer, which provides single output for diagnosing the rice disease. [10]

Kumar, Singh, Kumar, and P. Singh have proposed a method named the Crop Selection Method (CSM) to solve the crop selection problem and maximize the net yield rate of crops over a season and subsequently achieve maximum economic growth for the country [11]. Son et al. have developed a ML approach for predicting rice crop yields in Taiwan using time-series Moderate Resolution Imaging Spectroradiometer (MODIS) data. Random forests (RF) and SVM were used for building predictive models [12].

12.3 METHODOLOGY

The dataset being used for the study was processed using the WEKA tool. "WEKA (*Waikato Environment for Knowledge Analysis*) [13], is freely available and open source data mining tool available under the GNU General Public License." This tool was developed at the University of Waikato, New Zealand, and was first implemented in its modern form in 1997. In addition to the arff data file format, the WEKA tool also supports .csv data file format, which makes it convenient to export data into different spreadsheet applications. We have used WEKA version 3.8.6 to perform our experiments for the purpose of explanation.

The dataset that has been used as an input to classifiers is available on the Kaggle repository. The name of this dataset is the "crop recommendation" dataset. It is a multivariate dataset with 2,200 instances, and its size is 65 KB. The total number of attributes in this dataset is eight, including the target label attribute. This dataset is classified for different crops, but we have converted the label attribute of the dataset into yes/no according to rice crop recommendations. The selected attributes of the dataset under study are defined in Table 12.1.

12.3.1 Workflow

The input dataset is processed using two ML classifiers – Naive Bayes and SVM. These classifiers are trained and tested for accuracy using the same dataset. Based on the experimental results collected, the best performing

TABLE 12.1

Description of Attributes

S. No.	Attribute Used in Dataset	Type of Attribute	Description
1	N	Numeric	Ratio of nitrogen content in soil
2	P	Numeric	Ratio of phosphorous content in soil
3	K	Numeric	Ratio of potassium content in soil
4	Temperature	Numeric	Temperature in degrees Celsius
5	Humidity	Numeric	Relative humidity in percentage
6	Ph	Numeric	Ph value of the soil
7	Rainfall	Numeric	Rainfall in mm
8	Label	String	Yes/No

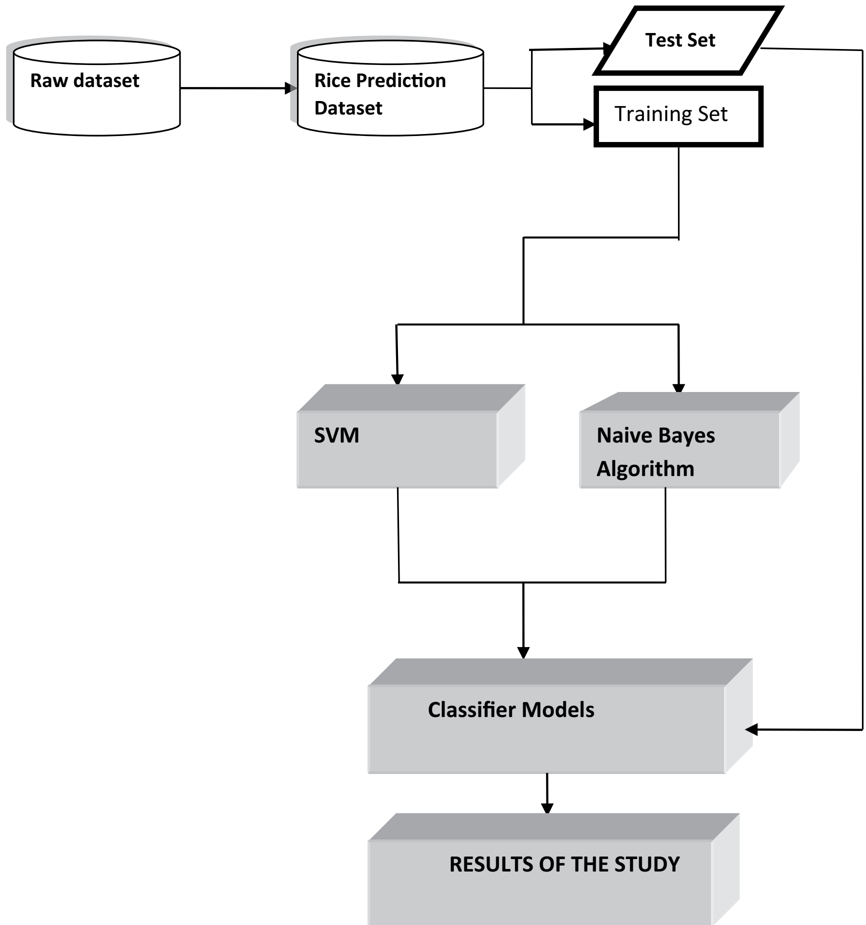

FIGURE 12.1
Flowchart of proposed methodology.

algorithm is determined that will help in the prediction of rice crops to grow on a particular farm based on the parameters depicted in Figure 12.1.

12.3.2 Procedure

The following procedure was followed while building the classifiers and calculating the accuracy of the algorithms using the WEKA tool.

1. The WEKA GUI chooser tool gives the flexibility to choose between Simple CLI, Knowledge Explorer, Experimenter, and Explorer CUI. The experiment was done using the WEKA Explorer interface.

2. Load the input dataset of rice crop recommendation in the WEKA tool.
3. Select the ML algorithm as Naive Bayes or SVM Sequential Minimal Optimization (SMO).
4. Cross-validation was selected as a test option, which is set as ten folds by default.
5. Test the classifier model for the selected algorithm.
6. Perform the comparison of the results obtained from the algorithms.

12.4 CLASSIFIERS EXPLAINED

12.4.1 Naive Bayes

Naive Bayes is a supervised learning classifier that works on the principle of conditional probability. With multiple inputs (x1, x2, x3, x4…xn), the outcome class (Y) using the Naive Bayes equation can be calculated as

$$P(Y=K \mid X_1 \ldots . X_n) = \frac{\begin{pmatrix} P(X_1 \mid Y=K) * P(X_2 \mid Y=K) * P(X_3 \mid Y=K) \\ * \ldots * P(X_n \mid Y=K) * P\left(Y=K\right) \end{pmatrix}}{\left(P\left(X_1\right) * P\left(X_2\right) * P\left(X_3\right) \ldots P\left(X_n\right)\right)}. \tag{12.1}$$

In the dataset under study, the Naive Bayes classifier (12.1) is applied to the following attributes:

X = (N, P, K, Temperature, Humidity, Ph, rainfall) and the outcome class Y is classified as Y = (Yes, No).

Different types of evaluation matrices are available in ML (e.g., "Classification Accuracy, Logarithmic Loss, Confusion Matrix, Area under Curve, F1 Score, Mean Absolute Error, and Mean Squared Error"). In this case study, we have considered the Confusion Matrix, Classification Accuracy, and Area under Curve (AUC) evaluation measures to do the comparative analysis of algorithms.

12.4.1.1 Naive Bayes Confusion Matrix

Table 12.2 shows the Confusion Matrix obtained by using the Naive Bayes classifier. Thus, according to this classifier, the values of True Positive = 100, False Negative = 0, False Positive = 43, and True Negative = 2057.

TABLE 12.2

Confusion Matrix Obtained by Using Naive Bayes

		Predicted Class	
Actual Class		Positive	Negative
	Positive	100	0
	Negative	43	2057

12.4.2 Support Vector Machines

SVM is a supervised learning algorithm, which is usually used for classification and regression-based problems. The primary concept of SVMs, initially developed for solving problems related to binary classifications, is the use of hyperplanes to define decision boundaries between data points of different classes. SVMs can handle both simple and linear classification tasks, and can also help in handling more complex – i.e., nonlinear – classification problems. Both separable and nonseparable problems can be well handled by SVMs in linear and nonlinear cases. The idea behind SVMs is to map the original data points from the input space to a high-dimensional, or even infinite-dimensional, feature space such that the classification problem becomes simpler in the feature space. The mapping is done by a suitable choice of a kernel function [14]. In WEKA, we have used a SMO algorithm for training a support vector classifier using a linear kernel on the given dataset in order to build the classification model.

K(x,y) = <x,y> and classifier for class True, False

12.4.2.1 Support Vector Confusion Matrix

Table 12.3 shows the Confusion Matrix obtained by using the support vector classifier. Thus according to this classifier, the values of True Positive = 64, False Negative = 36, False Positive = 3, and True Negative = 2097

TABLE 12.3

Confusion Matrix Obtained by Using SVM

		Predicted Class	
Actual Class		Positive	Negative
	Positive	64	36
	Negative	3	2097

TABLE 12.4

Accuracy Results for Classifiers

Algorithm	Time Taken to Build the Model (sec)	Accuracy Value (%)
Naive Bayes	0.01	98.0455
SVM using WEKA tool	0.11	98.2273

12.5 ACCURACY

Accuracy is another performance evaluation matrix for classification models (Table 12.4). It is calculated as the total number of true predicted instances to the total number of instances presented in the dataset.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}), \tag{12.2}$$

where in (12.2) TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

Figure 12.2 graphically depicts the accuracy results for the classifiers in which time taken to build the models and accuracy values are shown.

12.6 AREA UNDER CURVE

AUC is another way of measuring the performance of an ML model. The higher the AUC, the better the performance of the model at distinguishing

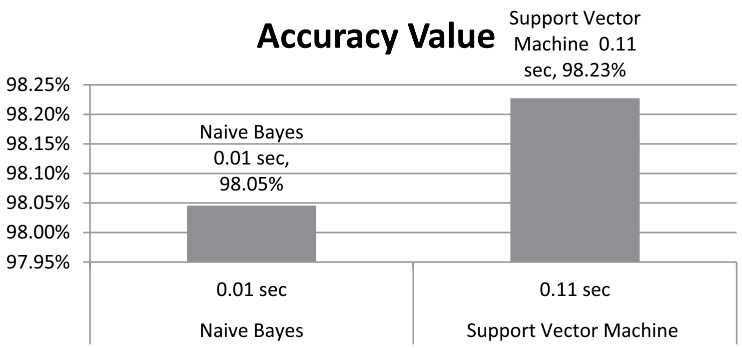

FIGURE 12.2

Graphical representation of accuracy results.

between the positive and negative classes. Following are the classifier accuracy measure values:

I. True-Positive Rate:
II. False-Positive Rate:
III. Precision:
IV. Recall:
V. F-Measure:

I. True-Positive Rate: It depicts what portion of the positive class gets correctly classified. It is calculated as

$$\text{TruePostive} - \text{Rate} = \frac{\text{TruePosive}}{\left(\text{TruePositive} + \text{FalseNegative}\right)}. \tag{12.3}$$

II. False-Positive Rate: It depicts what portion of data is classified as abnormal, which is actually normal.

$$\text{FalsePositive} - \text{Rate} = \text{FalsePositive} / \left(\text{FalsePositive} + \text{TrueNegative}\right) \tag{12.4}$$

III. Precision: It depicts the number of correct positive predictions made. It is calculated as

$$\text{Precision} = \frac{\text{TruePositive}}{\left(\text{TruePositive} + \text{FalsePositive}\right)}. \tag{12.5}$$

IV. Recall: It depicts the no. of correct positive predictions made out of all positive predictions that could have been made. It is calculated as

$$\text{Recall} = \frac{\text{TruePositive}}{\left(\text{TruePositive} + \text{FalseNegative}\right)}. \tag{12.6}$$

V. F-Measure: It depicts the overall performance of the classifier. It is calculated as

$$\text{F} - \text{Measure} = 2 * \left(\text{Precision} * \text{Recall}\right) / \left(\text{Precision} + \text{Recall}\right). \tag{12.7}$$

TABLE 12.5

AUC Measures of Algorithms

Algorithm	TP Rate	FP Rate	Precision	Recall	F-Measure
Naive Bayes	0.980	0.001	0.986	0.980	0.982
SVM	0.982	0.344	0.982	0.982	0.981

Table 12.5 represents AUC measures (12.3)–(12.7) of these algorithms obtained after building the classifiers on the dataset in the WEKA tool.

12.7 CONCLUSION

In this chapter, a prediction model based on Naive Bayes and SVM classifiers has been employed to predict whether a particular farm can be recommended for rice cultivation.

The dataset used in carrying out the study for the prediction of farms was obtained from the Kaggle repository, which contains different soil and environmental behavioral attributes.

Naive Bayes and SVM classifiers were built using the WEKA tool. These classifiers thus have been compared based on different accuracy measures to check the performance using the same dataset on two different models. The study led to the conclusion that the TP Ratify Rate and Recall of SVM are higher compared to Naive Bayes. However, the precision and F-measure of Naive Bayes are higher, as compared to SVM. The accuracy value of SVM is 98.2273%, which is slightly better as compared to the Naive Bayes classifier. However, the time taken in building the model in Naive Bayes is 0.01 sec, which is less compared to the time taken using the SVM. Since we should be mainly concerned about the accuracy of the model rather than the time taken in building the model, SVM is the best pick established by the current comparative study.

In the future, we can compare other prediction models like logistic regression, decision support trees, random forest, and neural networks with SVM to establish better prediction models for the current dataset. Moreover, the attribute set can also be enhanced based on various other environmental factors to establish more accurate results so that we may be able to select the right farm for the cultivation of rice crops.

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